

ROYAL FITNESS CLUB

Nutrition & Healthy Eating Certificate

MODULE 1

Macronutrients Deep Dive

Duration	60 minutes
Course Code	CS_B2
Level	Beginner
Format	Study Guide PDF

Part of the Royal Fitness Club Professional Certification Program.

SECTION 1: The Role of Macronutrients in Human Physiology

Beyond Calories — What Macros Actually Do

Macronutrients (protein, carbohydrates, and fats) are the three categories of large molecules that provide energy and structural material to the human body. While calories determine body weight over time, macronutrient composition determines body composition — the ratio of muscle to fat — and profoundly influences hormones, inflammation, satiety, cognitive function, and athletic performance.

A calorie is not just a calorie in terms of physiological effects. 500 kcal from chicken breast triggers different hormonal responses, satiety signals, and metabolic fates than 500 kcal from refined sugar — even though both are 500 kcal.

Macronutrient	Energy Density	Primary Functions	Dietary Sources
Protein	4 kcal/g	Muscle tissue, enzymes, hormones, antibodies	Chicken, fish, eggs, dairy, legumes, soy
Carbohydrates	4 kcal/g	Primary fuel for CNS and high-intensity exercise, glycogen storage	Grains, fruits, vegetables, pulses
Fats	9 kcal/g	Hormones, cell membranes, fat-soluble vitamins	Oils, nuts, seeds, fatty fish, avocado, dairy
Alcohol (reference)	7 kcal/g	No nutritional function; suppresses fat oxidation	Beer, wine, spirits — to be minimised

SECTION 2: Protein — A 360° View

The Science of Dietary Protein

Amino Acid Profiles and Essential vs Non-Essential

Proteins are polymers of 20 amino acids connected by peptide bonds. Of the 20, 9 are classified as **essential** (must be obtained from diet; cannot be synthesised): histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine.

The remaining 11 are **conditionally essential** (normally synthesisable but required from diet under illness, injury, or rapid growth): arginine, cysteine, glutamine, glycine, proline, serine, tyrosine, asparagine, aspartate, glutamate, alanine.

Complete vs Incomplete Proteins

Category	Definition	Examples	Limitation for Vegans
Complete proteins	Contain all 9 essential amino acids	Eggs, dairy, meat, fish, soy, quinoa	Most animal proteins are complete

Incomplete protein	Missing or low in one or more EAAs	Most plant proteins (grains, legumes, nuts)	Complementary combining needed
Complete protein	Contains all EAAs in sufficient amounts	Animal products (dairy, eggs, meat)	Classic staple in Indian/Latin American diets
Complementary proteins	Two incomplete proteins together = complete EAA profile	Example: rice + dal	

Leucine: The Anabolic Trigger

Among the 20 amino acids, leucine occupies a unique position: it directly activates mTORC1, the master regulator of protein synthesis. A minimum leucine threshold of approximately 2–3g per meal is required to maximally stimulate muscle protein synthesis (MPS).

Food	Protein per 100g	Leucine per 100g	Serves to Achieve 2g Leucine
Chicken breast	31g	2.5g	80g chicken
Whole eggs	13g	1.1g	~4 eggs (200g)
Whey protein powder	80g	9g	25g powder
Lentils (cooked)	9g	0.7g	285g lentils
Tofu (firm)	8g	0.6g	333g tofu
Paneer	18g	1.4g	143g paneer

This data explains why a 30g scoop of whey efficiently triggers MPS, while the same amount of a plant source may not — the leucine density differs. Vegans should target the higher end of protein intake (2.0–2.4g/kg) to compensate for lower leucine density.

Protein Metabolism: Digestion to Deposition

- Stomach: HCl denatures protein; pepsin hydrolyses peptide bonds
- Small intestine: pancreatic proteases (trypsin, chymotrypsin) complete breakdown to amino acids and small peptides
- Absorption: amino acids cross intestinal epithelium via specific transporters; enter portal circulation
- Liver: regulates amino acid release to peripheral tissues; urea cycle removes excess nitrogen
- Muscle: uptake driven by insulin and exercise; mTORC1 activation → ribosomal activity → protein synthesis

Protein Timing Revisited

Meal	Protein Target	Food Examples	Rationale
Breakfast	25–35g	3 eggs + 200g Greek yoghurt	Breaks overnight fast, MPS stimulus after 6–8h sleep
Lunch	25–40g	200g chicken / fish + 100g dals	Maintains MPS stimulus; leucine threshold met
Pre-workout (if 2h+ before)	20–30g	Paneer bowl, Greek yoghurt, Amaranth	Amino acids circulating during training

Post-workout	25–40g	Whey shake / chicken / eggs	Maximise post-exercise MPS window
Dinner	25–40g	Fish, chicken, dal + rice, tofu	Complete EAA profile at final meal
Pre-sleep	30–40g casein	Cottage cheese, casein supplement	Slow digesting; sustains MPS for 7–9h of sleep

SECTION 3: Carbohydrates — Structure, Function, and Optimisation

Carbohydrates: More Than Just Energy

Carbohydrate Digestion and Glycaemic Response

All digestible carbohydrates are ultimately converted to monosaccharides (primarily glucose) before absorption. Glucose enters the bloodstream, triggering insulin release from pancreatic beta cells. Insulin drives glucose into muscle (for glycogen synthesis), liver (glycogen), and adipose tissue (fat).

The rate of this process determines the glycaemic response — fast digestion = rapid glucose spike = rapid insulin spike = subsequent glucose dip (potential energy crash). Slow digestion = sustained glucose = sustained energy without excessive insulin spikes.

Dietary Fibre — The Overlooked Powerhouse

Dietary fibre is the non-digestible carbohydrate component of plant foods. Despite not contributing energy directly, fibre is arguably the most important carbohydrate for long-term health:

Fibre Type	Source	Health Effect
Soluble (viscous)	Oats (beta-glucan), legumes, apples, citrus	Lowers LDL cholesterol; slows glucose absorption; prebiotic
Insoluble	Wheat bran, vegetables, whole grains	Reduces colon transit time; reduces colorectal cancer risk
Resistant starch	Cooked-cooled rice/potato, unripe banana, legumes	Feeds beneficial gut bacteria to butyrate; prebiotic; glucose blunting
Prebiotic fibre (FOS, inulin)	Garlic, onion, chicory, asparagus, Jerusalem artichoke	Feeds beneficial Bifidobacterium; gut microbiome diversity

Target fibre intake: 25–38g/day (women at the lower end; men at the higher end). Most Indians eat 15–20g/day — a significant deficit. Dal (lentils), whole wheat roti, and sabzi (vegetables) are culturally accessible ways to reach the target.

Carbohydrate Loading and Periodisation

Strategic carbohydrate manipulation around training sessions can optimise performance and body composition simultaneously:

Strategy	When	Carb Amount	Goal
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Carb loading	24–72h pre-endurance event (>90min)	8–12g/kg BW/day	Maximise glycogen stores
High-carb training day	Before intense sessions	3–5g/kg	Performance; glycogen full
Low-carb training day	Before low-intensity / rest day	1–2g/kg	Enhance fat oxidation adaptation
Intra-workout carbs	During sessions >60 minutes	30–60g/hour	Maintain blood glucose
Post-workout carbs	Within 30–60min (after protein)	0.8–1.2g/kg	Rapid glycogen resynthesis

SECTION 4: Fats — The Complete Guide

Fat Is Not the Enemy

Essential Fatty Acids

Two polyunsaturated fatty acids cannot be synthesised by the human body and must be obtained from diet: linoleic acid (omega-6) and alpha-linolenic acid (ALA, omega-3). They are the "parents" of longer-chain derivatives critical for physiology:

Parent EFA	Derivatives	Sources	Primary Role
Linoleic acid (LA) — omega-6	Arachidonic acid (AA)	Sunflower, soybean, corn	Eicosanoid precursors — pro-inflammatory signalling (in excess)
Alpha-linolenic acid (ALA) — omega-3	EPA, DHA (conversion rate <10%)	Fatty fish, flaxseed, walnuts	Anti-inflammatory; brain DHA requires direct EPA/DHA from fish

The omega-6:omega-3 ratio problem: Modern diets have omega-6:omega-3 ratios of 15:1 to 20:1 (vs ancestral ~4:1). This chronic excess of omega-6 drives systemic low-grade inflammation linked to cardiovascular disease, obesity, depression, and inflammatory conditions. The solution: reduce refined seed oils (sunflower, soybean, corn) and increase EPA+DHA from fatty fish, fish oil, or algae oil supplements.

Cholesterol — Separating Science from Myth

Dietary cholesterol (found only in animal foods — eggs, shellfish, organ meats) has a minimal effect on blood cholesterol in most people because the liver compensates by producing less endogenous cholesterol when dietary intake increases. "Cholesterol responders" (25% of population) do see modest LDL increases with high dietary cholesterol, but context matters — egg consumption within a calorie-controlled diet does not increase cardiovascular risk in most studies.

Cholesterol Type	What it Is	Target	Influence
LDL-C (LDL cholesterol)	Cholesterol carried by LDL	<100 mg/dL optimal; <70 for high risk	Saturated fat increase LDL; MUFAs lower it
HDL-C (HDL cholesterol)	Cholesterol carried by HDL (reverse transport)	>60 mg/dL (men), >50 (women)	Exercise, moderate alcohol, mono/polyunsaturated fats increase
Triglycerides	Circulating fat in blood	<150 mg/dL	Excess refined carbs, sugar, alcohol raise TG; omega-3, exercise lower

LDL particle count (MLD) is a specific CVD risk marker that is less useful than LDL-C. Small, dense LDL particles are more atherogenic than large, buoyant LDL particles.

SECTION 5: Calculating and Tracking Macros

Practical Macro Setup

Setting up macros follows a hierarchy: total calories first, then protein, then fat (minimum for hormones), then fill remaining calories with carbohydrate.

Step-by-Step Macro Calculation Example

Example: 28-year-old female, 60kg, moderately active (1.55 multiplier), goal: body recomposition

Step	Calculation	Result
1. BMR (Mifflin-St Jeor)	$(10 \times 60) + (6.25 \times 163) - (5 \times 28) - 161 = 1380$ kcal	1380 kcal
2. TDEE	1380×1.55	2139 kcal
3. Caloric target (250 kcal deficit)	$2139 - 250$	1889 kcal
4. Protein (2.0g/kg)	$2.0 \times 60\text{kg} = 120\text{g} \times 4$ kcal	480 kcal
5. Fat (25% of TDEE)	$0.25 \times 1889 \div 9$ kcal/g	52g fat = 472 kcal
6. Carbohydrates (remaining)	$1889 - 480 - 472 = 937$ kcal $\div 4$	234g carbs
Summary	1889 kcal 120P / 234C / 52F	Balanced recomp macro set

Module 1 (CS_B2) Key Takeaways
• Macros provide energy but also structural, hormonal, and regulatory functions beyond calorie yield • Leucine threshold ~2–3g per meal required to maximally activate MPS via mTORC1 • Fibre (25–38g/day) improves glycaemic control, gut microbiome, satiety, and cardiovascular health • Omega-6:omega-3 ratio matters — reduce seed oils, increase EPA+DHA from fish/algae • Macro calculation: calories → protein (2g/kg) → fat (min 15–25%) → fill with carbs • Carbohydrate periodisation (high on training days, lower on rest days) optimises recomposition

1. What is the leucine threshold and why does it make plant proteins potentially inferior for MPS?
2. Explain resistant starch: how is it formed, and what are its health benefits?

3. Calculate macros for a 75kg male seeking muscle gain at 2800 kcal total.
4. Why is the omega-6:omega-3 ratio more important than total fat intake for inflammation?
5. Distinguish between LDL-C and LDL particle count (LDL-P) as cardiovascular risk markers.